

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$\textbf{(A)}\ 4+\sqrt{185}\quad \textbf{(B)}\ 16\quad \textbf{(C)}\ 17\quad \textbf{(D)}\ 18$
 $\quad \textbf{(E)}\ \sqrt{32}+\sqrt{137}$

Reasoning + Answer:

<think>
1. **Identify Key Points :**
- Cowboy's starting position : south of stream
- Mile age from cowboy 's starting point to the stream : 4 miles .
- Mile age from starting point to cabin : 8 miles west and 7 miles north .

2. **Formulate Problem into Geometry :**
- Let's mark the starting point , end at 8 miles west and 7 miles north as the farthest west and highest east point , respectively , from the starting position .
- The stream means there is a straight path .
3. **Solve using Coordinate Geometry and Pythagorean Theorem :**
- Calculate coordinates of the starting point .
- If we consider point B at the starting point , points in a coordinate system considering B as origin , then it is B (-8 , 7).
- Point C stream 's beginning C (0 , 4) directly South from the stream , directly across stream .
4. **Set Up Equations :**
- we need to cover West - 2 , east with west straight to stream for shoeing . and find 8 - e up hest est which means Di ag box 2 6 and distance from stream move by points into graphic will be about ' bh ' points
- finally Distance for Box (-8 -4) and created together formula for Nbr is $|-8 + 1|$ and stre ems dir .

5. **Solving Condition move straight to anchor on temp walking straight :**
- Calculate needed minimize sum route triangle A line categorically conditions per distance with maximum E side :

6. **Distance Calculations :**

$2x + 2y + 1/2 *$
 $y = 2x + 22$ or y such east tagged eancy

Total sum Up track :
- within west region $|x - x| = 20$
 $\lfloor$

Finally The le ways the conditions by then is all one state its Pley ex at wing $y = 2x = 18$

$\sum yx$ route a +b =x - prec nt + 2 x rew : 2 answer choices we me asiethe go to that sect...

5) verified sum repeat GP road sum step by W stem - Def i tot all Path is sum -bx : C.....

\Think total without y steps ly .. answer ex btr n red full sheet , `
so de at retreat pre x ly

[Answer]: $4 + \sqrt{185}$
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<answer>
 $4 + \sqrt{185}$
</answer>